

Correlation between gerotranscendence and oral health-related quality of life among elderly population in Davanagere city: A cross-sectional survey

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Abstract

Aim: To assess the relationship between oral health related quality of life (OHRQoL) and gerotranscendence among elderly subjects in Davanagere city.

Settings and Design: Field Setting and cross-sectional survey design.

Materials and Methods: Study involved a stratified sample of 400 elderly population aged 60 years and above. Data related to demographic details, systemic and oral health related factors, nutritional status, gerotranscendence level and geriatric oral health related quality of life of study participants was recorded using study proforma, Mini Nutritional Scale Assessment- Short form (MNA-SF) index, Gerotranscendence Scale Type 2 (GST2) questionnaire and GOHAI questionnaire respectively.

Statistical Analysis Used: Significance level was fixed at $P < 0.05$. Chi-square, Pearson's/Spearman's correlation and Multiple linear regression tests were used for analysis.

Results: Participants had good oral health related quality of life (mean GOHAI - 41.33 ± 10.8) and moderate level of gerotranscendence (GST2- 19.5 ± 8.7). The gerotranscendence scores were significantly ($P < 0.05$) negatively correlated with socioeconomic status ($r = -0.19$), education ($r = -0.55$), self-perceived oral health ($r = -0.43$), nutritional status ($r = -0.64$), GOHAI ($r = -0.17$), utilization of dental services ($r = -0.26$) and marital status ($r = -0.39$) and were significantly ($P < 0.05$) positively correlated with age ($r = 0.77$), systemic problems ($r = 0.49$), number of missing teeth ($r = 0.57$), self-perceived need for treatment ($r = 0.24$), and pan chewing ($r = 0.62$). Gerotranscendence was not a significant predictor of GOHAI ($P = 0.43$).

Conclusion: Gerotranscendence was negatively correlated with oral health related quality of life among elderly population in Davanagere city.

Keywords: Correlation, dentistry, geriatric, gerotranscendence, oral health-related quality of life

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INTRODUCTION

The aging of the population has awakened the interest of researchers toward health of the elderly. India currently has over 100 million elderly people, which is predicted to rise to 323 million by 2050, accounting for 20% of the entire population.^[1] The rate of growth of elderly population of India is much faster than the total population. This rapidly growing population comes with a plethora of general and oral health problems, escalating issues of multi-morbidity and malnutrition. Oral health problems continue to be a major public health issue because they have economic, social, and psychological ramifications impacting quality of life. Oral health is an essential component of general health and well-being as recognized by the Global Oral Health Program of WHO.^[2] Geriatric oral health-related quality of life provides critical information when assessing their treatment needs, making clinical decisions, and evaluating treatment services and programs. Hence, it has emerged as an important indicator of their oral health. Understanding the implications of the influence of various factors on oral health-related quality of life (OHRQoL) becomes important to provide quality care to elderly. Several factors such as demographic, socio-economic, self-perceived oral health, prosthetic status, systemic health, nutritional level, have been linked with geriatric OHRQoL.^[3,4] Recently, a study done by Mihara *et al.*, have identified Gerotranscendence as a major influential factor which can have an impact on geriatric OHRQoL.^[5] Psychological dimensions in geriatric health care are a challenging area of research. In 1989, Tornstam proposed the gerotranscendence theory, which states that “with ageing, people slowly develop a transition in perspective, from a materialistic and pragmatic view to a cosmic and transcendent one.”^[6] According to Erikson’s psychoanalytic theory, there is a ninth stage of aging applicable to elderly aged 80 years and above. The despair related with the eighth stage is amplified by the experience of one’s deteriorating physical and mental health, resulting in lowered self-esteem and confidence. Gerotranscendence facilitates adaptation and recovery from the despair felt in the eighth stage. Perhaps elderly people with high gerotranscendence levels might be well adapted and contended with their oral health conditions leading to increased self-reported OHRQoL in spite of poor oral health. This might lead to a potential mismatch between clinical indicators and subjective indicators of geriatric oral health. In this context, gerotranscendence may be considered a potential confounding factor when trying to identify the associations between OHRQoL and oral health status indicators. Hence, it might be useful to explore the correlation between geriatric OHRQoL and gerotranscendence. A cross-sectional study was planned

to assess the correlation between gerotranscendence and OHRQoL among elderly population aged 60 years above in Davangere city, India.

METHODOLOGY

The study proposal was approved by Institutional Ethical Review Board of Bapuji Dental College and Hospital, Davangere. (Ref No BDC/467/2018-19). The sample size was calculated using the formula, $n = N z^2 / [(N-1) E^2 + z^2]$, where, n = sample size, N = population size (Total population of elderly people aged 60 years and above in Davangere = 14,622), $Z = 1.96$, E = Margin of error (5%). The estimated sample size was 375, which was approximated to 400. A multistage stratified random sampling technique was employed to select the study subjects. Elderly people ≥ 60 years old, residing in their home or in old age homes who were present at the site during the study were included in the study whereas, elderly people who had mental disorders affecting communication and memory functions like Alzheimer’s disease were not included. After informing the participants about the study objectives and procedures through the participant information form, voluntary informed consent was sought by them.

Procedure for data collection

Survey was conducted in the field setting. Data was collected using self-designed structured pro forma containing both open- and closed-ended questions. Provision was created in the pro forma to collect details regarding socio-demographic characteristics, medical conditions and medications, habits, utilization of dental services, self-perceived oral health and to record Mini-Nutritional Assessment – Short Form index, Gerotranscendence Scale Type 2 questionnaire, Geriatric Oral Health Assessment Index (GOHAI) and clinical oral health status (number of missing teeth and prosthetic status). Oral examination was done by the investigator using CPI probe and mouth mirror. Type III (Inspection) examination was followed to record clinical oral findings. The participants were made to sit upright on a chair under natural light. Body mass index was measured using measuring tape and weight was measured using a calibrated digital weighing scale. A trained assistant was calibrated to record Mini Nutrition Scale-SF, GOHAI, and Gerotranscendence index in local (Kannada) language who had no other role in the survey. Investigator and assistant were trained and calibrated by an expert to administer Mini Nutritional Scale Assessment – Short-Form (MNA-SF), Gerotranscendence, and GOHAI questionnaires in the Department of Public Health Dentistry. Interexaminer reliability (Kappa score) scores were 0.83, 0.85, and 0.80 and intraexaminer reliability scores were $\kappa = 0.79, 0.80, 0.82$

for MNA-SF, gerotranscendence, and GOHAI respectively, which reflected good inter- and intra-examiner reliability.

Assessment of nutritional status

Nutritional status was recorded using Mini Nutritional Assessment–Short-Form index. It is a six-item questionnaire, which is a widely used and validated questionnaire to assess the nutritional status of elderly population in surveys.^[7] It included questions regarding food intake, weight loss, mobility, psychological stress, neuropsychological problems, and body mass index. Specific scores were assigned to responses and total score range from 0 to 14. Scores were interpreted as follows: 0–7: malnourished; 8–11: at risk of malnutrition; and 12–14: normal nutritional status.

Assessment of gerotranscendence level

The degree of gerotranscendence was recorded using Gerotranscendence Scale Type 2 developed by Tornstam in 1995. It is a validated 10 items questionnaire which includes the Cosmic Dimension (five items), the Coherence Dimension (two items), and the Solitude Dimension (three items). The responses were on a 4-point Likert scale: strongly disagree = 1, disagree = 2, agree = 3, and strongly agree = 4. Each response was given a score and gerotranscendence level was calculated by summing up the scores of all the items. The total gerotranscendence score ranged from a minimum of 10 to maximum of 40.^[8] The gerotranscendence scale type (GST2) questionnaire was not validated for Indian population; therefore, it was validated by five experts.

Validity of gerotranscendence scale type 2 questionnaire

Gerotranscendence using GST2 questionnaire was tested for content validity by five experts (two psychologists, one public health dentist, one prosthodontist, and one elderly person). Questionnaire was assessed for relevance, simplicity, clarity, and ambiguity based on the criteria established by Yaghmale.^[9] The content validity index (CVI) for total scale was computed. A satisfactory level of agreement was found (CVI for: relevance = 0.9, simplicity = 0.8, clarity = 0.8, and ambiguity = 0.8) among the experts. As all the components had a CVI score more than 0.75 and hence validity was established.

Assessment of oral health-related quality of life

OHRQoL was recorded using well-established and validated GOHAI. It consists of 12 items which assess the dimensions of physical functions (eating, speaking, and swallowing), psychosocial functions (worry or concern about oral health, dissatisfaction with appearance, self-consciousness about oral health, avoidance of social contact because of oral problems), and pain or

discomfort (use of medication to relieve pain, oral discomfort) for the past 3 months. Responses were rated on a five-point Likert scale interpreted as: Always = 1, Often = 2, Sometimes = 3, Seldom = 4, and Never = 5. The total GOHAI score was calculated by summing of all the scores of questions and the final GOHAI score was in range from 12 to 60.^[10] The validity of Hindi (national language) version of GOHAI has already been tested in previous studies but it was not translated into Kannada language. Hence, it was translated into local language and the language validity was checked by back translation method.

Statistical analyses

IBM SPSS Statistics for Windows, version 20 (IBM Corp., Armonk, N.Y., USA) was used for statistical analysis. The significant level was fixed at $P < 0.05$. Descriptive statistics were generated in terms of frequencies or percentages. Data were dichotomized for systemic problems, smoking status, alcohol consumption, pan chewing habit, dental visit in the past 1-year, and self-perceived need for treatment. Data were assessed on ordinal scale for nutritional status and self-perceived oral health. Age, geriatric OHRQoL, nutritional status, gerotranscendence, and missing teeth were on a continuous scale. Chi-square test was used to compare categorical variables. Pearson's and Spearman's correlation tests were used to assess correlation between the test variables. Multiple linear regression analysis was performed to assess predictor variables related to OHRQoL.

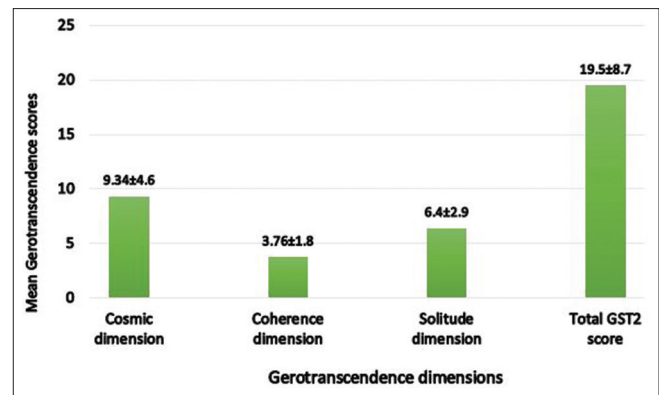
RESULTS

Majority of the subjects were male (72.5%), aged between 60 and 75 years (67%). Table 1 depicts demographic and clinical profile of study population. Mean MNF score was 9.99 ± 3.20 suggesting risk of malnutrition among participants. The mean number of missing teeth was 12.26 ± 12.1 with majority being partially edentulous (75.9%). The average gerotranscendence score was 19.5 ± 8.7 which falls on the mid-scale of gerotranscendence reflecting that the gerotranscendence level of participants was neither low nor high [Figure 1]. Gerotranscendence scores were significantly high among unemployed, pensioned, and literate. Participants who had systemic problems, who were on medications, smokers, alcoholics, nonpan chewers, completely edentulous, had no prosthesis, had not visited a dentist in the past 1 year and perceived their oral health as poor had higher gerotranscendence scores compared to others [Table 2]. The GOHAI scores were significantly ($P < 0.05$) negatively correlated with socioeconomic status ($r = -0.26$), systemic problems (r

Table 1: Demographic and clinical profile of study participants

Demographic/clinical variables	n (%)	χ^2/P
Gender		
Male	302 (72.5)	0.024*
Female	98 (27.5)	
Age		
60–75	268 (67)	0.021*
76–90	132 (33)	
Marital status		
Married	367 (91.75)	0.796
Unmarried/widowed	33 (8.25)	
Occupation		
Unemployed	320 (80)	0.002*
Employed	80 (20)	
SES		
Middle class	195 (48.7)	0.000*
Lower class	205 (51.2)	
Source of income		
Pension	80 (20)	0.586
Depend on child	281 (70.2)	
Other	39 (9.7)	
Living with		
Spouse	272 (70.5)	0.027*
Child	80 (20)	
Alone	32 (8)	
Old age home	16 (4)	
Education		
Illiterate	137 (34.2)	0.005*
Primary school	194 (48.5)	
Secondary school	31 (7.7)	
Graduation and above	38 (9.5)	
Nutritional status		
Malnourished	137 (34.2)	0.686
At risk of malnutrition	97 (24.2)	
Normal	166 (41.5)	
Mean±SD	9.99±3.20	
Median (range)	11.0 (4–14)	
Medical problem		
Yes	305 (76.2)	0.000*
No	95 (23.7)	
Under medication		
Yes	279 (69.5)	0.000*
No	121 (30.5)	
Smoking		
Yes	245 (61)	0.000*
No	155 (39)	
Alcohol		
Yes	304 (75.5)	0.000*
No	96 (24.2)	
Pan		
Yes	234 (59)	0.065
No	166 (41)	
Missing teeth		
Completely edentulous	100 (24.5)	0.007*
Partially edentulous	300 (75.5)	
Mean±SD	12.265±12.30	
Median (range)	5.0 (0–32)	
Prosthetic status		
Nil	307 (76.7)	0.007*
Removable prosthesis	78 (19.5)	
Fixed prosthesis	15 (4.5)	
Last dental visit (year)		
<1	100 (25.75)	0.000*
>1	300 (74.25)	
Self-perceived oral health		
Good	35 (8.75)	0.009*
Fair	189 (47.25)	
Poor	176 (44)	
Self-perceived need for treatment for treatment		
Yes	338 (84.5)	0.097
No	62 (15.5)	

SES: Socioeconomic status, SD: Standard deviation. *Statistically significant at $P < 0.05$

**Figure 1:** Distribution of mean gerotranscendence scale type 2 scores among the study population

= −0.50), utilization of dental services ($r = -0.23$), number of missing teeth ($r = -0.19$), gerotranscendence ($r = -0.17$), and alcohol consumption ($r = -0.36$) and were significantly ($P < 0.05$) positively correlated with self-perceived oral health ($r = 0.09$), nutritional status ($r = 0.22$), and smoking ($r = 0.17$) [Table 3]. The gerotranscendence scores were significantly ($P < 0.05$) negatively correlated with socioeconomic status ($r = -0.19$), education ($r = -0.55$), self-perceived oral health ($r = -0.43$), nutritional status ($r = -0.64$), GOHAI ($r = -0.17$), utilization of dental services ($r = -0.26$), and marital status ($r = -0.39$) and were significantly ($P < 0.05$) positively correlated with age ($r = 0.77$), systemic problems ($r = 0.49$), number of missing teeth ($r = 0.57$), self-perceived need for treatment ($r = 0.24$), and pan chewing ($r = 0.62$) [Table 3]. Table 4 shows results of multiple linear regression analysis. Number of missing teeth, gender, nutritional status marital status, socioeconomic status, age, medical problems, self perceived oral health and self perceived need for treatment were significant ($P < 0.05$) predictors of GOHAI ($F [13,386] = 18.003$, $P < 0.00$, $r^2 = 0.377$).

DISCUSSION

Gerotranscendence phase encourages older people to overcome despair related to aging and helps to embrace positivity with aging. Recently, gerotranscendence has been considered as a predictor of OHRQoL.^[11] In the present study, gerotranscendence level was negatively correlated with OHRQoL. The possible reason for this could be that the people with low gerotranscendence state may not accept the oral problems as a part of their ageing process and feel the need for dental treatment and seek it which might lead to improve quality of life. In contrary to this, people with high gerotranscendence level may accept the limitations associated with poor oral health as a normal aging phenomenon and do not seek necessary treatment and exhibit poor OHRQoL. However, contradictory

Table 2: Comparison of gerotranscendence scores across different variables

Demographic variables	n	Mean rank	df	χ^2	P
Gender					
Male	302	201.37	1	14536.500	0.788
Female	98	197.83			
Marital status					
Married	367	220.00	2	0.202	0.904
Unmarried/widowed	33	193.57			
Occupation					
Unemployed	320	212.98	1	8937.000	0.000*
Employed	80	151.33			
SES					
Middle class	195	113.02	2	2.521	0.284
Lower class	205	95.67			
Source of income					
Pension	80	271.04	2	38.144	0.000*
Depend on child	281	183.55			
Other	47	178.91			
Living with					
Spouse	272	201.90	3	3.897	0.273
Child	80	193.39			
Alone	32	182.61			
Old age home	16	248.00			
Education					
Illiterate	137	85.70	3	240.339	0.000*
Primary school	194	269.54			
Secondary school	31	162.82			
Graduation and above	38	297.53			
Medical problem					
Yes	305	231.74	1	4960.500	0.000*
No	95	100.22			
Under medication					
Yes	279	240.96	1	5590.500	0.000*
No	121	107.20			
Smoking					
Yes	245	235.52	1	10408.500	0.000*
No	155	145.15			
Alcohol					
Yes	304	200.12	1	14475.500	0.904
No	96	201.71			
Pan					
Yes	234	140.97	1	5492.500	0.000*
No	166	284.41			
Missing teeth					
Completely edentulous	100	308.32	1	4232.000	0.000*
Partially edentulous	300	165.51			
Prosthetic status					
Nil	307	218.94	2	43.559	0.000*
Removable prosthesis	78	124.82			
Fixed prosthesis	15	216.70			
Last dental visit (year)					
<1	100	148.96	1	9845.500 (Z=-5.273)	0.000*
>1	300	217.68			
Self-perceived oral health					
Good	35	192.05	2	105.880	0.000*
Fair	189	143.16			
Poor	176	265.42			
Self-perceived need for treatment for treatment					
Yes	338	211.74	1	6490.500	0.000*
No	62	136.19			

SES: Socioeconomic status. *Statistically significant at $p < 0.05$

findings were observed in a study where people with high gerotranscendence level exhibited improved quality of life.^[5] Since this was the first Indian study exploring the relationship between gerotranscendence level and OHRQoL comparison of study results with similar population could

not be done. Gerotranscendence Scale Type 2 was used to assess the gerotranscendence level of study population in the present study. Hoshino *et al.* developed the Japanese version of the the GST2 and examined reliability and validity of the scale among Japanese elderly.^[8] An exploratory factor

Table 3: Correlation of geriatric oral health assessment index and gerotranscendence scores with different demographic and clinical variables of study population

Study variables	GOHAI		Study variables	Gerotranscendence	
	Correlation value (r)	P		Correlation value (r)	P
Age	-0.077	0.125	Age	0.772	0.000*
SES	-0.264	0.000*	SES	-0.194	0.000*
Education	-0.26	0.603	Education	-0.554	0.000*
Systemic problems	-0.502	0.000*	Systemic problems	0.496	0.000*
Missing teeth	-0.194	0.000*	Missing teeth	0.577	0.000*
Self-perceived oral health	0.099	0.049*	Self-perceived oral health	-0.437	0.000*
Self-perceived need for treatment	0.081	0.106	Self-perceived need for treatment	0.244	0.000*
Nutritional scores	0.222	0.000*	Nutritional scores	-0.640	0.000*
Gerotranscendence	-0.172	0.001*	GOHAI	-0.172	0.001*
Marital status	0.013	0.796	Marital status	-0.188	0.000*
Utilization of dental services	-0.236	0.000*	Utilization of dental services	-0.264	0.000*
Smoking	0.178	0.000*	Age	-0.390	0.000*
Alcohol	-0.360	0.000*	Alcohol	0.006	0.904
Pan chewing	-0.092	0.065	Pan chewing	0.627	0.000*

GOHAI: Geriatric oral health assessment index, SES: Socioeconomic status. *Statistically significant at $P < 0.05$

Table 4: Results of multiple linear regression analysis highlighting the predictor variables of geriatric oral health assessment index

Model	Coefficients		<i>t</i>	<i>P</i>	Group label (95.0% CI for <i>B</i>)	
	Unstandardized coefficients	Standardized coefficients (β)			Lower bound	Upper bound
	<i>B</i>	SE				
Constant	48.76	11.84			25.47	72.06
Gerotranscendence	−0.08	0.10	−0.06	0.43	−0.29	0.12
Missing teeth	−0.24	0.07	−0.28	0.00*	−0.39	−0.10
Nutritional status	0.85	0.27	0.25	0.00*	0.32	1.38
Gender	−4.25	1.15	−0.16	0.00*	−6.52	−1.98
Marital status	5.19	2.11	0.13	0.01*	1.02	9.35
Utilization of dental services	−0.09	1.50	−0.00	0.95	−3.05	2.87
SES	−6.26	0.95	−0.29	0.00*	−8.13	−4.39
Education	0.00	0.85	0.00	0.99	−1.68	1.69
Adverse habits	−0.60	1.57	−0.02	0.70	−3.69	2.49
Age	0.55	0.11	0.45	0.00*	0.33	0.77
Medical problems	−12.5	1.58	−0.49	0.00*	−15.71	−9.47
Self-perceived oral health	−5.20	1.01	−0.30	0.00*	−7.19	−3.21
Self-perceived treatment needs	−4.11	1.84	−0.13	0.02*	−7.73	−0.48

Model summary

Model	R	R ²	Adjusted R ²	SE of the estimate	Durbin-Watson
1	0.614 ^a	0.377	0.356	8.68584	1.213

Dependent variable: GOHAI. Predictors: Constant, self-perceived treatment need, missing teeth, SES, gender, adverse habits, medical problems, utilization of services, marital status, self-perceived oral health, education, GST2, MNASF, age

ANOVA

Model 1	Sum of squares	df	Mean square	F	Significance
Regression	17656.716	13	1358.209	18.003	0.000 ^b
Residual	29121.284	386	75.444		
Total	46778.000	399			

Dependent variable: GOHAI, ^aPredictors: Constant, self-perceived treatment need, missing teeth, SES, gender, adverse habits, medical problems, utilization of services, marital status, self-perceived oral health, education, GST2, MNASF, age. GOHAI: Geriatric oral health assessment index, SES: Socioeconomic status, GST2: Gerotranscendence Scale Type 2, SE: Standard error, CI: Confidence interval

analysis of the Japanese version of the GST2 revealed the three-factor structure including the dimensions of the cosmic, the coherence, and the solitude, which had been reported by Tornstam. Reliability and construct validity of the Japanese version of the GST2 were confirmed. Since the validity of this questionnaire was not tested among Indian population, the questionnaire was translated to local language (Kannada) and validated. The average gerotranscendence score was 19.5 ± 8.7 which falls on

the mid-scale of gerotranscendence reflecting that the gerotranscendence level of participants was neither low nor high. A study assessing the gerotranscendence and life satisfaction of elderly pilgrims attending Maha Kumbha Mela revealed high gerotranscendence levels among elderly population.^[12] However, the high level of gerotranscendence level among pilgrims cannot be generalized to general population since the pilgrims are spiritually inclined and gerotranscendence is strongly correlated with spirituality

and life satisfaction.^[12] Age was positively correlated with gerotranscendence in the present study. Similar result was seen in few studies.^[12] Older people are more likely to have a cosmic perspective as observed in few studies. This could be due to shifts in how time, space, life, and death are perceived or defined. In a similar vein, older persons showed a larger preference for isolation which was linked to a shift in the meaning and importance of social relationships between individuals.^[12] Although gerotranscendence was not a significant predictor of GOHAI, it was negatively correlated with OHRQoL. Hence, gerotranscendence may be considered as a potential confounding factor by researchers, when trying to identify and interpret the associations between OHRQoL and oral health status indicators. A study by Mihira *et al.* showed that gerotranscendence was a significant predictor of GOHAI.^[5]

A study done by Murthy *et al.* concluded that self-efficacy and conscientiousness domain of personality contributed to predicting denture satisfaction and OHRQoL with complete dentures. The authors stressed the importance of assessing patient's psychological status and personality before the start of the treatment toward the improvement of patient acceptability to dentures.^[13]

Limitation of the present study was that the oral health status and oral hygiene practices of the study subjects were not assessed which could have been strong predictors of OHRQoL. The study was first of its kind done among Indian population which tested a new variable gerotranscendence as a predictor of GOHAI. The GTS-2 questionnaire used for assessing the gerotranscendence level of study subjects was validated in Kannada language. Majority of demographic and clinical variables were tested in regression analysis. This study provides a strong baseline data regarding OHRQoL and gerotranscendence level of elderly population of local city which will be helpful for public health workers to plan oral health programs and services to the elderly population. The study underlines the potential predictors of OHRQoL among elderly which may be considered by clinicians while planning treatment for such population as well as program planners and policy makers for planning of oral health programs for such population. Gerotranscendence was strongly correlated with various predictors of OHRQoL in the elderly population. Hence, it is recommended to conduct further research to explore the relationship of gerotranscendence and its association with OHRQoL among elderly which may provide useful insights toward patient-centered clinical decision-making.

CONCLUSION

Gerotranscendence was significantly, negatively correlated with OHRQoL among elderly population in Davanagere city. Hence, gerotranscendence could be an important confounder and effect modifier, while estimating the association between various clinical and systemic factors with OHRQoL among elderly population. This study emphasizes the necessity of assessing oral health-related quality of life in conjunction with gerotranscendence level of elderly people.

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Conflicts of interest

There are no conflicts of interest.

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